

EXPERT INTERVIEW

Objective: To understand clinically meaningful behavioral patterns associated with Major Depressive Disorder and how they reflect treatment progression.

Abstract: This research investigates the application of artificial intelligence–driven digital phenotyping to model treatment response trajectories in individuals receiving care for Major Depressive Disorder (MDD). The study utilizes passively collected behavioral and physiological data from mobile and wearable sensing technologies, including measures of sleep regularity, activity levels, mobility, routine stability, mobile device interaction patterns, and autonomic signals. The objective is to characterize longitudinal changes that correspond to clinically meaningful progression, including symptom improvement, sustained remission, stagnation, or early relapse risk.

Building on established evidence linking behavioral indicators to depressive symptom dynamics, this work advances treatment monitoring by analyzing temporal patterns and intra-individual variability across the course of care. Machine learning–based approaches are employed to detect shifts in behavioral trajectories that may signal significant changes in clinical status, with emphasis on modeling progression rather than static symptom severity.

Recognizing the constraints associated with accessing real-world clinical datasets, the study incorporates the development of a structured synthetic dataset grounded in clinically plausible behavioral trajectories. This framework integrates variables with established or theoretically supported relevance to depressive symptomatology and functional functioning, enabling methodological development in a privacy-preserving environment. Clinical expertise is integrated to ensure alignment between the modeled behavioral patterns and therapeutic reasoning.

The overarching aim is to integrate computational modeling with clinical insight, supporting privacy-sensitive and ecologically valid approaches to treatment response monitoring in MDD.

Expert Interview Guide: Clinically Meaningful Digital Behavioral Patterns in MDD Treatment Progression

Opening:

Thank you for taking the time to speak with me. I'm conducting this interview to better understand which behavioral changes clinicians consider meaningful when monitoring treatment progression in Major Depressive Disorder (MDD). The purpose is to ensure that the behavioral variables I model in my research are clinically grounded and aligned with therapeutic reasoning.

Consent:

Your participation is voluntary, and you may skip any question or stop the interview at any time. The discussion is focused on professional expertise rather than patient-identifiable information. With your permission, I would like to take notes and audio-record to ensure accuracy. The information will be used for research purposes and reported in an anonymized way. Do I have your consent to proceed?

Interview focus:

This interview explores which behavioral changes clinicians consider meaningful in treatment progression for MDD and how these changes could be reflected in digital behavioral data collected between clinical sessions.

Section 1: Clinically Meaningful Behavioral Change

Primary question:

When a patient with MDD is responding to treatment, what behavioral changes do you typically observe that indicate improvement, and what changes would raise concern for deterioration or relapse?

Probing prompts:

1. Do you focus more on changes relative to an individual's baseline, or on absolute levels of behavior and functioning?
2. When behavior changes during treatment, what patterns most strongly suggest improvement versus early relapse risk?
3. Which behavioral changes tend to emerge early in recovery, and which occur later?
4. Are there behavioral changes that might look like improvement in digital data but could reflect distress or worsening clinically?
5. Do you place more weight on a single behavioral change, or on several changes occurring together, when judging clinical significance?

Mapping prompt:

If you were to translate the changes you described into measurable digital signals, what behaviors would be most important to capture?

Section 2: Temporal Patterns and Variability

Primary question:

When assessing treatment progression in MDD, how do you interpret short-term day-to-day changes compared to longer-term trends over weeks or months?

Probing prompts:

1. Over what time frame do you typically interpret change as clinically meaningful (days, weeks, months)?
2. Which is more informative clinically: short-term fluctuations or sustained patterns over time?
3. Are patterns of stability versus instability in daily life clinically meaningful, and in what way?
4. What patterns over time suggest improvement or recovery, and what patterns raise concern for worsening?

(sustained deterioration, increasing variability, or abrupt change)

Mapping prompt:

What kinds of patterns over time would you want to see in digital behavioral data?

Section 3: Functional Recovery vs Symptom Severity

Primary question:

How do you distinguish symptom improvement from functional recovery during treatment for MDD, and what real-world changes matter most clinically?

Probing prompts:

1. Can self-reported improvement occur without functional recovery (or vice versa)?
2. Which domains of functioning are most informative clinically?
3. How do you interpret mismatch between self-report and observed functioning?
4. Under what conditions is increased engagement in daily life a meaningful sign of recovery?

Mapping prompt:

Which observable behaviors best represent functional recovery between sessions?

Section 4: Early Warning Signals of Relapse

Primary question:

What early behavioral or physiological warning signs tend to appear before relapse or symptom worsening becomes clearly apparent?

Probing prompts:

1. How far in advance do these warning signs typically emerge?
2. Do sleep/routine disruptions usually precede mood decline or follow it?
3. Is social withdrawal gradual or abrupt?
4. Are physiological stress-related changes clinically meaningful or mainly supportive?
5. Which early signs are most reliable across patients?

Mapping prompt:

If you could track only a few early warning behaviors between sessions, what would you prioritize and why?